

RESPONSE

from Prof. Marco Porta (University of Pavia)
on the diploma project on the topic
"Classification of manual assembly phases using
multimodal signals from eye and head movement tracking"
of the student

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The work of Georgi Chaporov focused on classifying human assembly-line actions using eye- and head-tracking data. The NEON wearable eye tracker was employed to capture multimodal signals from participants as they built LEGO sets, simulating a manual manufacturing environment. A custom Python software was developed for labeling data and managing video overlays of gaze points.

Several challenging tasks were addressed in the work, among which were:

- Achieving generalization across different participants (model accuracy dropped significantly from subject-dependent trials to subject-independent trials);
- Classifying highly subjective behaviors (such as the "Searching for pieces" phase);
- Synchronizing and temporally aligning multimodal sensor data (eye-tracking, 3D eye-state, and IMU head-movement data), as the sensors operated at different sampling frequencies;
- Handling activity label boundaries (finding the exact start and end times).

The work was successfully carried out with great maturity and autonomy, and certainly meets the requirements for a thesis in Computer Science or Engineering. The carried out study produced the following main results:

- Assembly phases were classified through deep learning models, with up to 76.87% accuracy for known subjects (temporal sequences proved to be essential for successful classification);
- The CNN-LSTM-Fusion model was the most effective architecture for generalizing to unseen subjects;
- "Assembling" was the easiest phase to recognize, while "Searching for pieces" was highly subject-dependent.

I propose that the student Georgi Chaporov be admitted to the diploma defense.
I rate his work as excellent.

Supervisor:

/ Marco Porta /

04.09.2026

